Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	On-Device Deployment of Large Language Models via Algorithmic Optimization			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		*Information & Communication*	
	Medium Category	Small Category	Medium Category	Small Category
	Information Security	*Industrial Security and Convergence Security*	*Software*	*Software Solutions*
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The proposed research will develop a comprehensive suite of technologies that enable efficient deployment of Large-Language models (LLMs) and Vision-Language models (VLMs) on resource-constrained edge devices including smartphones, tablets, augmented reality headsets, and embedded systems. The proposed research will have a broad range of applications including industrial automation, healthcare and video surveillance. It will also directly aim to improve trustworthiness of AI systems by developing practical approaches that can be deployed on edge devices at consumer premises rather than relying on cloud-based solutions. The core concept in the proposed research is to improve efficiency of large language (LLMs) and vision (LVMs) models through hardware-aware optimization techniques. We will focus on three main directions: (1) knowledge distillation methods from large model to smaller models (2) speculative decoding and (3) adaptive quantization. Collectively these technologies will improve the efficiency existing LLMs and LVMs enabling their deployments on resource constrained edge devices without a significant loss in model accuracy.			

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Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* The PI and his research team have been actively engaged in conducting cutting edge research on improving model efficiency of LLMs. The proposed R&D project provides a valuable opportunity to steer our prior research towards real world deployments. The PI has engaged in discussions with Nota AI, a Korean SME that is an emerging industry leader in hardware aware optimization of LLMs and LVMs to enable deployment on resource constrained edge devices. Through these discussions, we have identified key topics for collaborations based on mutual interest. As elaborated in this document, the PI and his research team will develop new algorithmic solutions involving knowledge distillation, speculative decoding and adaptive quantization for system models directly relevant to the business needs of Nota AI. This LOI has been jointly written by the PI and the R&D team at Nota AI. If the proposed research is successful, it will expand the PI's research program to incorporate new directions such as LVMs and lead to high impact research. The technology developed in the proposed research will also be integrated in the industry partner's products such as the NetsPresso platform and lead to commercialization opportunities.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* *1. Model Efficiency of Large Language Models* 2. Knowledge Distillation 3. Speculative Decoding 4. Model Quantization 5. Model Optimization

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (1-2)	End of the Project	Level (6)

R&D Objectives

Describe the specific technologies you aim to develop through this R&D project

Current state-of-the-art LLMs and VLMs require billions of parameters and substantial computational resources, confining their deployment to cloud-based infrastructure. This creates fundamental limitations around latency, and cost that constrain real-world applicability in deployment over edge devices. This project aims to overcome these limitations through principled compression and optimization techniques that preserve model capability while reducing resource requirements. The research project will involve three technologies that provide complementary approaches to improve efficiency of LLM and VLM architectures:

1. Topic1: Cross-Tokenizer based Knowledge Distillation
2. Topic2: Speculative Decoding
3. Topic3: Adaptive Quantization

In each of these topics the PI and his team will build upon their recent works [1-4] and extend these methods to algin with the business needs of Nota AI by developing new algorithmic solutions. Each of the three technologies will be pursued in parallel during this project. In the final year, these our approaches will be integrated into a single solution that will be optimized for speed, memory requirements and computational cost to enable deployment over resource constrained devices.

R&D Target Specifications

Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters

Topic1: Cross-tokenizer Knowledge Distillation

This approach will enable efficient transfer of knowledge between VLMs employing different tokenization schemes for both visual and textual modalities. Classical distillation approaches [8] work best when the teacher and student models to share identical tokenizers. When the models share different tokenizers the training methods are very heuristic and incur performance degradations [9, 10]. This severely limits architectural flexibility and prevents optimization of tokenization strategies for different deployment scenarios. In our recent research [1] we have proposed a principled approach for computing likelihood scores for LLMs when the tokenizers for the teacher and student models differ. This has resulted in significant improvements in the performance of the student model. Our proposed research will further build upon this work and extend it to incorporate applications to VLMs trained across different tokenizers.

The main R&D target specifications are as follows:

(a) Vocabulary Size Reduction: Support student vocabularies from 16K to 64K tokens while teacher uses 100K+ tokens

(b) Use different visual tokenizers for teacher and student models

(c) Retain at-least 90% of the accuracy of teacher model upon distillation.

Topic 2: Speculative Decoding

Autoregressive generation, where each token is generated sequentially conditioning on all previous tokens, is inherently slow because it prevents parallelization across the time dimension and is memory-bandwidth bound rather than compute-bound. Speculative decoding addresses this by using small, fast draft models to generate candidate sequences that are verified in parallel by the large target model. Classical approaches to speculative decoding [5] can provide up to a factor of 2 improvement in the token rate over standard autoregressive decoding. Speculative decoding has

become an active area of research in the past few years and there have been several innovations on sampling draft tokens [11-13], draft selection [14, 15] and multi-draft extensions [16, 17].

Most existing methods involve certain dependence between the output token sequence and the drafter --- for the same random seed upon changing the drafter the actual LLM output can change. The is not a desirable property in practice as the output should not be sensitive to the choice of the draft model. Our work in [2] presents a novel speculative decoding method that supports drafter invariance and avoids such dependence. Our method is considerably simpler than the classical approaches [2] and in fact achieves state-of-the-art token rates over baselines in the setting when multiple drafts are used. The proposed research will build upon this prior work and incorporate extensions to Vision Language Models (VLMs).

The main target specifications in the proposed research are as follows:

(a) Study the ideal choice of draft models that incorporate both vision and language tokens in VLMs
(b) Achieve block efficiency of 2.5 to 3.5 tokens accepted per draft model invocation.
(c) Drafter Invariance: Demonstrate Identical Distributions even when draft distributions are changed.

Topic 3: Quantization of Large Language Models

The third topic will focus on quantization techniques specifically designed for large language models and vision language models. Quantization represents model parameters and activations with reduced-precision numerical formats, enabling dramatic reductions in memory footprint and computational requirements. In this project we will focus on post-training quantization (PTQ) schemes where we have access to a pre-trained LLM/VLM with full precision weights and quantize it to a lower precision representation without significant loss in accuracy. This setting is of most practical relevance as the actual training of LLMs/VLMs involves a significant computational burden. The literature on post-training quantization of LLMs has evolved significantly in the past few years [18, 19, 20].

A critical innovation in the proposed research is dynamic bit-width adaptation that enables runtime adjustment of quantization levels. In contrast most of the current approaches that target a fixed bit-width i.e., the weights are quantized to just one precision level. The proposed technology will employ nested codebook structures that enable efficient conversion between different bit-widths. In our prior work [4], we have proposed such a quantization strategy that represents each weight as a sum of codewords from multiple codebooks. This method enables transition from 8-bit to 6-bit representations by dropping certain codewords rather than re-quantizing from full precision. This allows a single model to support multiple bit-width providing flexibility to adapt to varying resource constraints without storing multiple versions. This is particularly attractive for dynamic hardware aware optimization of LLMs as the model weights can be selected at different precisions during run-time based on device constraints and model sensitivity. The proposed research will build upon this initial work to study the suitability of this approach to a broader class of LLMs and VLMs. We will aim for the following target specifications:

(a) Develop adaptive/nested quantization schemes that support at-least 3 different bit-widths. Benchmark the performance against ideal quantization schemes that target each bit-width separately and demonstrate that the penalty associated with the adaptive scheme is small.
(b) Evaluate the proposed quantization scheme in terms of storage and training time and demonstrate a significant improvement (e.g. 50% reduction in storage and training time) over traditional methods that produce a separate model for each bit-width.
(c) Evaluate the improvements from mixed-precision quantization over single-precision quantization in the context of VLMs. Explicitly identify which modules require higher precision quantization and which modules can support lower precision quantization without affecting the accuracy over a broad range of LLM tasks.

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R&D Plan

Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives

Topic1: Cross-tokenizer Knowledge Distillation

Methodology:

We will initially handle the case when the teacher and student apply an identical vision mode l for generating visual tokens but use different language models with different tokenizers. In t his setting the cross-tokenizer likelihood computation from our recent work in [1] is directly a pplicable. We will perform experiments that demonstrate the benefits of near-exact likelihood scoring methods in knowledge distillation for VLMs. We will consider practical settings where the student model deploys 16K to 64K tokens while teacher uses 100K+ tokens. Our next st ep will be to generalize the likelihood conversion approach to the setting where the teacher and student models use different vision encoders. In such settings exact computation is gene rally not possible and approximate likelihood mappings are computed by mapping the encode r outputs to shared space representations [6]. We will improve upon such heuristics by levera ging upon the exact conversion algorithms for language models [1]. We will perform extensiv e experiments to demonstrate the improvements of current baseline distillation methods.

Milestones and Timeline:

- Year 1: Develop Cross tokenizer likelihood method when teacher and student models share the same visual encoders
- Year 2: Extend the above method to incorporate the setting when the teacher and st udent models use different visual encoders
- Year 3: Compare with existing baselines and quantify the advantages of the proposed methods.

Topic 2: Speculative Decoding

Methodology:
To achieve maximums speedups in the token rates from speculative decoding, a close alignment between the draft and target models is a key requirement. Our first step will be to study the architectures of draft models that provides closest alignment with the target models. This will involve appropriate choice of both vision encoders (if any) and language models that on one hand have faster execution and on the other hand have close alignment with the target models. We will particularly focus on the speculative decoding scheme in our prior work [2]. The proposed method has lower complexity that standard speculative decoding and supports draft invariance i.e. the output tokens generated are identical regardless of the choice of the drafter when the random seed is fixed. This is particularly important in real world systems, as the dependence of the output on the choice of the drafter can be undesirable in practice. Since our proposed method [2] also generalizes naturally to multiple drafters, we will also consider how multiple drafters can be defined in the context of VLMs. As one example, different drafters could be conditioned on different objects in the input image/video and the speculative decoding algorithm will dynamically select the drafter that is most relevant to the given query.

Milestones and Timeline

- Year 1: Develop an architecture for speculative decoding for VLMs based on different candidate choice of draft models and study the tradeoff between execution time and alignment with the target models
- Year 2: Extend the speculative decoding algorithm in [2] to VLM settings ands study t he performance gains in single draft and multi-draft settings
- Year 3: Perform extensive experiments to compare with existing baselines and demon strate grains under real world constraints.

(max 10 pages)

Topic 3: Quantization of Large Language Models

Methodology:

We will primarily focus on post-training quantization (PTQ) where the goal is to quantize the weights of a pre-trained LLM/VLM. Such methods are practically useful as training an LLM/VLM requires significant resources and is not a viable solution in most cases. We will focus on a specific quantization scheme called Additive Quantization for Language Models (AQLM) where the quantized weights are representation as a summation of constituent codebooks [7]. This methods is particularly effective for lower bit-widths. It also has a nested structure so that one can achieve different bit-width representations by simply dropping certain codebooks in the representation. Our preliminary work has studied this approach for two different bit-widths – 8 bit and 6 bit representations. Our proposed work will extend to approach to span multiple-different bitwidths. We will also extend the loss function so that it accounts for the choice of different bit-widths. Our aim will be to achieve the performance at each stage to match the performance of an ideal scheme where a single model is designed to target just the single bit-width representation. In the context of VLMs we will particularly investigate the use of mixed-precision quantization where different levels of precision can be applied to different layers as well as the vision encoder, language decoder and adaptation module.

Milestones and Timeline:

- Year 1: Extend AQLM to support multiple bit-widths for LLMs. Develop appropriate loss function that incorporates the contribution of each stage.
- Year 2: Extend the above approach to VLMs. Study the effect of mixed-precision quantization for the vision encoder, the language decoder and the alignment modules.
- Year 3: Perform extensive experiments to compare with existing baselines and demonstrate grains under real world constraints.

Topic 4: Integration and Final Testing

Our final step will be to integrate the approaches developed in Topics 1, 2 and 3 into a single system and evaluate the overall performance improvements in the integrated solution. Our approach will be as follows:

- Cross-Tokenizer Likelihood Scoring (topic 1) will be used to train a smaller student model with a smaller language vocabulary and vision encoder that is strongly aligned with the teacher model.
- If the accuracy of the of the student model on different downstream tasks needs to be improved, we will apply the speculative decoding module from topic 2. Speculative decoding will guarantee lossless performance i.e. the output token stream will follow the distribution of the target model. We will also
- To further improve the token rates and reduce memory requirements we will apply quantization techniques to both the draft model and the target model outlined in topic 3.

Milestones and Timeline:
The integration effort will be conducted in Year 3 and the source code for the solution will be developed for potential integration with the industry partner. We will conduct experiments on a broad range of datasets to demonstrate the overall improvement in the token rates and the model accuracy over a broad range of downstream tasks.

Team Composition and University:

The team at University of Toronto will consist of the PI (Ashish Khisti) and two PhD students who have prior experience with model quantization, speculative decoding and multimodal LLMs. Additional research personnel may also be recruited as necessary to support timely completion of the proposed research. The team at Nota AI will consists of engineers and team leads who have a significant experience with on-device implementation of LLMs. The PI and

	his students are affiliated with Vector Institute of Artificial Intelligence and have significant access to compute resources managed by the Vector Institute (https://docs.alliancecan.ca/wiki/Killarney) which will be crucial for the success of the proposed research. It should be noted that University of Toronto is the top ranked university in Canda and among top 20 institutes worldwide. It has 13 Nobel laureates including Geoffrey Hinton, the Godfather of AI, the 2024 Nobel laureate in Physics. The cost of a PhD student at U of T is approximately $30k (CDN) which about 1/3rd the cost of a student at a comparable university in USA or Europe. Also it has a friendly IP policy and a lower overhead costs than comparable universities in USA and Europe.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* The research institute will be responsible for the **core research, and academic validation** of advanced model optimization and inference methodologies, with a primary focus on **automatic mixed-precision quantization and Vision-Language Model (VLM) efficiency**. **Mixed Precision Quantization** - Designing **hardware-aware automatic mixed-precision quantization algorithms** that dynamically adapt precision levels based on device constraints and model sensitivity. - Developing **generalizable algorithmic frameworks** that can be applied across multiple model architectures and heterogeneous hardware platforms. - Providing **reference implementations, experimental analyses, and comprehensive technical documentation** to support joint development and technology transfer. **VLM-Specific Research Scope** - **Methodology Development:** Conducting research on **drafter-invariant speculative decoding methods** for large Vision-Language Models, as well as **draft-model quantization approaches** *(optional)* to further improve inference efficiency. - **Academic Validation:** Establishing **official benchmarks and evaluation protocols** to rigorously assess proposed methods prior to validation on industrial benchmarks.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* The Korean SME partner will focus on practical implementation, industrial validation, and commercialization-oriented activities, translating joint research outcomes into deployable and market-ready solutions. Its main responsibilities include: - Integrating the proposed algorithms into real-world deployment pipelines and target hardware platforms. - Conducting device-level testing and performance validation under practical application scenarios. - Testing and validating proposed methods using industrial benchmarks developed by Nota. - Deploying the proposed methodologies into industrial products, including LVLM-based AI solutions. Supporting pilot applications and exploring commercialization opportunities based on the joint R&D outcomes.

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* **Expected Performance and System-Level Benefits** • Achieve **over 70% improvement in token generation throughput** without loss of model accuracy. • Enhance the **reliability and trustworthiness of AI systems**, leading to improved user experience in real-world applications. • **Prompt response:** Reduced generation latency narrows the time gap between real-world events and VLM inference results in industrial scenarios. • **Dense context processing:** Faster token generation enables richer and more detailed visual and contextual understanding. • **Drafter-invariant speculative decoding** is expected to significantly reduce **model maintenance and operational costs** across multiple LVLM variants. • Enable **task- and domain-invariant adaptation** of speculative decoding methodologies. **Technological Outcomes** • Development of a **novel automatic mixed-precision quantization algorithm** that autonomously adapts precision levels based on hardware capabilities and model sensitivity. • Establishment of a **generalized, hardware-aware quantization framework** supporting multiple devices and diverse model architectures without device-specific manual optimization. • Demonstrated **quantitative improvements in inference efficiency**, including reduced latency, lower memory footprint, and improved energy efficiency, while preserving model accuracy. **Commercialization Potential** • Enable **faster, more cost-effective deployment** of AI models on edge and embedded devices, lowering the adoption barrier for SMEs. • Provide a **core optimization technology** that can be integrated into AI deployment toolchains, SDKs, and optimization platforms. **Industrial and Economic Impact** • Strengthen the **global competitiveness of Korean SMEs** within AI and semiconductor ecosystems through differentiated, hardware-aware optimization technologies. • Promote **scalable and sustainable AI deployment** across industries such as smart manufacturing, intelligent devices, and edge AI services. • Contribute to a **robust international collaboration model** that bridges advanced academic research with real-world industrial applications.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization